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Cognitive Distance in Document Submission Systems

Why Users Struggle with Simple Digital Tasks

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Abstract

Document submission workflows are common interaction patterns in contemporary information systems, including public services, finance, healthcare, and digital platforms. Despite their apparent simplicity, these workflows often generate confusion, repeated failure, and task abandonment. Existing usability frameworks often interpret such outcomes primarily in terms of complexity, insufficient skill, or inadequate training. Although these explanations remain important, they do not fully account for breakdowns that persist even in systems designed to reduce apparent cognitive load.

This paper examines document submission systems through the lens of cognitive distance, defined here as the degree of separation between a system's conceptual structure and the mental model constructed by the user during interaction. Building on prior work that conceptualizes and provisionally operationalizes cognitive distance, the paper develops a qualitative analytical case centered on file-upload interactions. The analysis illustrates how routine design choices, including file-format restrictions, file-size constraints, and ambiguous terminology, can produce systematic misalignment between system logic and user understanding. The paper proposes the analytic term externalized processing to describe a recurring pattern in which systems shift reasonably automatable preprocessing tasks, such as format conversion and file compression, to users. This shift increases procedural effort and introduces additional interpretive burden, often forcing users to rely on external tools, trial-and-error strategies, or informal workarounds to complete otherwise ordinary tasks.

The paper argues that many breakdowns in document submission workflows, as reconstructed here, are not merely incidental design defects but can reflect a broader problem in how preprocessing work is distributed between users and systems. The analysis suggests that reducing cognitive distance requires reallocating appropriate preprocessing functions back to systems, improving conceptual transparency, and aligning system behavior more closely with user reasoning. The contribution is analytical rather than experimental: it extends cognitive distance into a concrete interaction domain and identifies externalized processing as a recurring allocation pattern through which system-side constraints become user-side preprocessing accompanied by interpretive burden.

Keywords: cognitive distance, usability, accessibility, human–computer interaction, document submission systems, file upload workflows, cognitive load, mental models, conceptual alignment, cognitive accessibility, externalized processing, interface comprehension

1 Introduction

Information systems increasingly mediate access to essential services, including identity verification, financial transactions, healthcare interactions, and administrative processes. As reliance on these systems grows, so does the expectation that they support interaction that is both efficient and understandable across diverse user populations.

One of the most common interaction patterns within these systems is document submission. Users are routinely asked to provide digital representations of official records, such as identification documents, proof of address, or supporting evidence for transactions. From a technical perspective, the interaction appears straightforward; from the user's perspective, it often proves otherwise.

Users are often required to upload files in specific formats, comply with strict file-size limitations, and interpret system terminology without explicit guidance. These requirements introduce friction that is frequently underestimated during system design [5]. The result is a recurring pattern of confusion, repeated attempts, reliance on external tools, and, in some cases, task abandonment.

Traditional usability frameworks often interpret these outcomes primarily as consequences of complexity or user error. Cognitive load theory, for instance, emphasizes the mental effort required to perform tasks and has guided many efforts to simplify interfaces [6]. Broader usability traditions likewise emphasize efficiency, feedback, and ease of use in interaction design [3, 4, 7]. While valuable, these perspectives do not fully explain why users struggle with interactions that appear simple and require relatively little visible effort, nor do they explicitly specify how systems redistribute preprocessing and interpretive responsibilities to users.

Prior work has introduced cognitive distance as a complementary concept for addressing this gap [1]. Cognitive distance refers to the degree of separation between a system's internal logic and the mental model a user constructs during interaction. The concept shifts the analytical focus from effort to interpretability, emphasizing the alignment between how systems operate and how users understand them.

Against this conceptual background, the present paper treats document submission systems as an analytically tractable case through which cognitive distance can be examined in practice. More specifically, it asks how cognitive distance emerges in routine upload interactions, how it can be analyzed using a recently proposed analytical framework [2], and what that analysis reveals about broader design patterns in information systems. The paper makes two contributions. First, it shows that seemingly simple upload workflows may remain difficult to interpret not only because they require effort, but because they redistribute preprocessing and interpretive responsibilities to users. Second, it defines externalized processing as the recurring allocation pattern through which system-side constraints are converted into user-side preprocessing and downstream interpretive breakdowns. While existing usability and workload frameworks describe inefficiency, error, and difficulty, they do not explicitly model how systems systematically relocate preprocessing and interpretive work to users as a distinct design-level mechanism.

2 Theoretical Background

2.1 Cognitive Distance and Interpretability

Cognitive distance refers to the degree of separation between a system's conceptual structure and the mental model constructed by the user during interaction [1]. The construct becomes especially salient when users can complete tasks successfully while lacking a coherent explanation of how or why those tasks produce specific outcomes.

Unlike cognitive load, which emphasizes the quantity of mental effort required to perform a task, cognitive distance captures the quality of understanding that emerges from interaction [6]. A system may therefore require only modest effort while remaining difficult to interpret, particularly when its internal logic is not aligned with user expectations or prior knowledge.

The two constructs may covary, but they are analytically distinct [1, 6]. Cognitive load concerns how many mental resources a task demands; cognitive distance concerns how far system logic remains from the user's understanding. A task may be demanding yet conceptually transparent, as in a long but well-explained workflow whose rules can be anticipated at each step. Conversely, a task may require little visible effort yet remain conceptually opaque, as when a single upload action triggers an unexplained rejection. The argument of this paper therefore does not rest on the claim that document submission systems are difficult simply because they require more work. It rests on the claim that such systems often require users to act without being able to infer the causal structure that determines success or failure.

Cognitive distance is therefore not reducible to task difficulty or interface complexity. It reflects a relational property between systems and users, shaped by how system behavior is presented, how feedback is interpreted, and how users construct meaning over time. As this distance increases, users are more likely to rely on memorization, trial-and-error strategies, or external assistance rather than developing stable, transferable understanding.

In this sense, cognitive distance provides a complementary lens for analyzing usability and accessibility, focusing not on whether users can act, but on whether they can make sense of their actions.

More formally, cognitive distance is adjacent to, but distinct from, three established lines of work. Cognitive load theory addresses the mental resources required for task performance [6]; usability frameworks evaluate interaction in terms such as effectiveness, efficiency, and clarity [3, 4, 7]; and mental-model research examines the representations users form about how systems work. Cognitive distance does not replace these perspectives. Rather, it specifies the degree of misalignment between system structure and the model a user can construct during interaction, making interpretability itself the central analytical concern. This perspective is also consistent with prior work that emphasizes greater directness and transparency in interaction design [3, 4, 7].

2.2 Operationalizing Cognitive Distance

Recent work has proposed an analytical framework for assessing cognitive distance through observable dimensions [2]. These dimensions include perceived understanding, predictive accuracy, interaction confidence, and conceptual transparency. In the present paper, they are used as analytic dimensions rather than as independently validated scales.

Perceived understanding refers to the user's ability to explain system behavior in their own words. Predictive accuracy captures the extent to which users can anticipate outcomes before acting. Interaction confidence reflects the stability of user decisions during interaction. Conceptual transparency refers to the degree to which the system itself makes causal links between inputs, rules, and outcomes explicit. Unlike predictive accuracy and perceived understanding, it concerns what the interface reveals, not only what the user can infer or report.

These dimensions are not direct measures of effort, time-on-task, or subjective workload. Rather, they assess whether users can form an intelligible model of system behavior. This distinction is important because a user may report high effort while still accurately understanding the system, or may complete a task quickly while remaining unable to explain why the system behaved as it did.

Together, these dimensions provide a structured way to analyze whether users not only complete tasks, but also understand them.

For analytical clarity, Table 1 summarizes the dissociation between cognitive load and cognitive distance across the four logically possible combinations. The table is offered as an analytical heuristic appropriate to a qualitative case paper, not as a claim of experimental measurement.

Table 1. Analytical dissociation between cognitive load and cognitive distance

	Low cognitive distance	**High cognitive distance**
Low cognitive load	A brief, predictable interaction whose logic is readily understandable. *Example:* a one-click confirmation with clear feedback and obvious consequences.	An interaction that is easy to perform but difficult to explain. *Example:* a single upload or automated decision that is accepted or rejected without intelligible reasons.
High cognitive load	A demanding task that remains conceptually transparent. *Example:* a long workflow whose steps are clearly explained and predictable.	A demanding task whose underlying logic remains opaque. *Example:* a document submission workflow requiring conversion or compression under poorly explained acceptance rules.

3 Methodological Approach

This study adopts a qualitative conceptual case analysis rather than an empirical case study. Rather than collecting new observational or experimental data, it develops a theoretically informed analysis of a commonly encountered interaction pattern through the lens of cognitive distance.

Document submission workflows were selected as the focal case for three reasons. First, they are common across domains and affect a wide range of users. Second, their structure is sufficiently consistent across implementations to support analytical transferability at the level of mechanism. Third, they expose multiple layers of implicit knowledge that users are often expected to possess.

The analysis reconstructs a generalized user interaction path and identifies points at which cognitive distance emerges. Each stage of the workflow is examined in relation to the operational dimensions of cognitive distance. The evidence in this non-empirical analysis is interpretive rather than observational: the reconstructed workflow, the constraints it presupposes, and the kinds of explanations or breakdowns those constraints make available to users. The aim is not statistical generalization but theoretical elaboration: to make a familiar interaction pattern analytically visible, to clarify the mechanisms that produce interpretive breakdowns, and to specify a conceptually plausible relationship between cognitive load and cognitive distance. Accordingly, the paper does not claim prevalence or causal magnitude; it specifies an interpretive mechanism and its implications for analysis and design.

4 Case Analysis: Document Submission Systems

4.1 Task Structure and Implicit Requirements

A typical document submission workflow involves several steps: the user must obtain or capture a document, ensure that it is in an accepted file format, reduce its size if necessary, locate the file on the device, and submit it through the system interface.

From the system's perspective, these steps are encoded as prerequisites; from the user's perspective, they often appear as a sequence of implicit challenges. Each step assumes knowledge that may not be present. Systems implicitly assume that users understand file formats, conversion processes, and compression techniques without explicit support.

Figure 1 summarizes this generalized workflow and highlights representative points at which cognitive distance tends to emerge. The figure is schematic rather than empirical: it does not claim that every

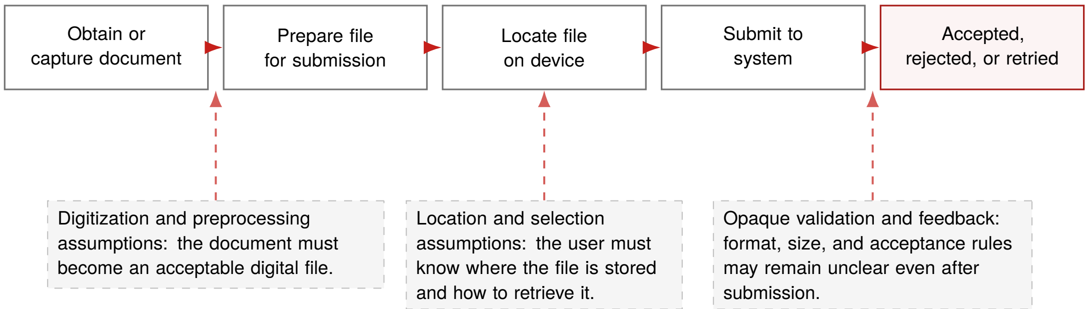

Figure 1. Generalized document submission workflow and representative points at which cognitive distance can emerge.

submission system follows exactly the same sequence, but it captures recurring interpretive demands that structure many upload tasks.

This expectation creates a gap between what the system requires and what the user can infer. The gap may remain partially invisible, yet it decisively shapes the interaction experience.

4.2 Externalization of Computational Tasks

A common workaround is to turn to external tools. Users search for ways to convert files between formats, use third-party applications to compress images, or otherwise modify inputs outside the system to make them acceptable. This observable behavior can be described as user-side preprocessing [8].

These behaviors suggest that some transformation tasks that are reasonably automatable within the system's technical, legal, and security context are instead offloaded onto users. Externalized processing does not refer to the necessary user input required to specify intent, supply source material, or authorize submission. It refers more narrowly to avoidable transformation work that the system could reasonably absorb but does not. Rather than adapting to user input, the system primarily filters it.

This distinction has direct implications for cognitive distance. Systems that process input internally can reduce, but not eliminate, the need for users to infer hidden transformation rules. By contrast, systems that enforce strict constraints without explanation or support increase the likelihood that users must guess how acceptable input is defined.

4.3 Analysis Across Cognitive Distance Dimensions

Applying the operational framework clarifies how cognitive distance manifests in this context.

In the terms of Table 1, the generalized workflow analyzed here most often occupies the high-load/high-distance quadrant. That placement, however, should not be reduced to effort alone. The same workflow could move toward high-load/low-distance if its requirements were clearly explained and supported within the interface, while a more automated workflow could still remain low-load/high-distance if its acceptance criteria stayed opaque.

Perceived understanding is limited when users cannot explain why certain files are accepted and others are rejected. Predictive accuracy declines when users must repeatedly attempt uploads without knowing whether the system will accept the file. Interaction confidence weakens as users rely on trial and error rather than on informed decision making. Conceptual transparency remains low when system constraints are presented without explanation or guidance.

Taken together, these conditions are consistent with elevated cognitive distance, even when users eventually complete the task.

5 Externalized Processing

5.1 Externalized Processing and Its Relationship to Cognitive Load

The case analysis suggests a recurring structural pattern in which preprocessing tasks are shifted from systems to users. This paper provisionally uses the term externalized processing for that mechanism. The associated observable behavior is user-side preprocessing, while *constraint without accommodation* names the interaction pattern in which systems specify valid input conditions without providing internal means of meeting them. The term is not intended to name a wholly separate psychological construct; rather, it isolates a design arrangement in which preprocessing work and the need to interpret system constraints are displaced onto users. The contribution is therefore not to relabel usability friction in general, but to identify a recurring allocation pattern through which design decisions about where transformation work occurs may generate downstream interpretive breakdowns. Although the pattern may increase cognitive load, it is not reducible to cognitive load alone.

Cognitive load refers to the mental effort required to perform a task. In document submission workflows, that effort becomes visible when users must convert file formats, reduce file size, or locate files within a device. These actions require attention, memory, and procedural execution, thereby increasing the demands placed on the user. Added steps, friction, and user work are therefore not identical to cognitive load, even though they may contribute to it by increasing mental processing demands.

However, the breakdowns reconstructed in document submission workflows cannot be reduced to effort alone. In many cases, users experience the task not only as demanding but also as unclear. The issue is not merely that the required steps demand cognitive effort; it is that they may not be immediately intelligible.

For example, when a system requires a file in a specific format, such as JPG or PNG, it implicitly assumes that the user understands what a file format is, how formats differ, and how to convert between them. Similarly, when a system enforces a file size limit, it assumes that the user understands what file size represents and how it can be modified. These assumptions introduce a form of misalignment that precedes effort. Before the user can decide whether to perform the required action, the user must first understand what the action means.

This distinction highlights the difference between cognitive load and cognitive distance. Cognitive load increases when users understand what is required but must expend effort to complete it. Cognitive distance increases when users cannot form a coherent interpretation of what is required in the first place. In such cases, additional effort does not necessarily resolve the problem, because the user lacks the conceptual basis needed to act effectively.

Externalized processing can amplify both conditions, albeit in different ways. By shifting preprocessing tasks to users, systems increase cognitive load by adding steps that require execution. At the same time, they can increase cognitive distance by exposing system-level concepts, such as file formats and compression techniques, that may not align with user mental models.

Figure 2 contrasts this user-side preprocessing arrangement with a system-side accommodation model. The comparison is conceptual rather than empirical, but it clarifies how the allocation of computational work changes both procedural effort and interpretive burden.

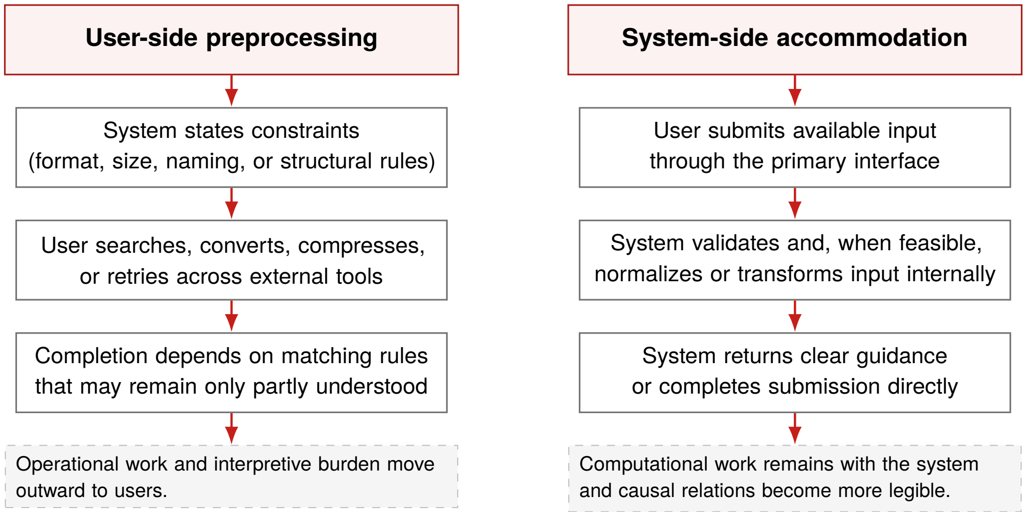

Figure 2. User-side preprocessing contrasted with system-side accommodation. Left: externalized processing shifts preprocessing and interpretive burden to users. Right: system-side accommodation retains feasible transformation and explanation within the system.

A related interaction pattern, referred to here as *constraint without accommodation*, arises when systems specify acceptable conditions for input but do not provide internal mechanisms to transform user-provided input into those conditions. As a result, users must not only perform additional work but also interpret system constraints that may remain conceptually opaque.

This distinction becomes especially relevant when user diversity is considered. Users with prior technical experience may experience the interaction primarily as increased cognitive load: they understand what is required but must invest additional effort to complete it. Other users, especially those with less technical experience or less familiarity with file-management concepts, may experience the same interaction primarily as high cognitive distance. For these users, the central problem is not effort but interpretation; they do not know what the system is asking, how to proceed, or how to recover from failure.

In this sense, cognitive load and cognitive distance may co-occur, but they are not equivalent. Cognitive load concerns the cost of performing understood actions. Cognitive distance concerns the gap between system requirements and user understanding. Externalized processing can increase both, while disproportionately affecting users who lack the conceptual resources needed to bridge that gap.

5.2 Theoretical Proposition

The following proposition is offered as a theoretical proposition derived from the conceptual case analysis above, not as a tested hypothesis. Here, externalized processing refers to reasonably automatable transformation work beyond the user input necessary to specify intent or provide source material.

Proposition 1: Information systems that externalize reasonably automatable preprocessing tasks to users, without preserving conceptual transparency through explanation or built-in assistance, are likely to exhibit higher levels of cognitive distance, as reflected in reduced predictive accuracy, lower interaction confidence, and diminished conceptual transparency.

This relationship is proposed as likely to hold across domains in which users must transform inputs to satisfy system constraints without receiving adequate interpretive support. As reliance on user-side preprocessing increases, alignment between system behavior and user understanding is likely to weaken, increasing reliance on trial-and-error strategies and external assistance. Future empirical work is needed to test the scope and strength of this proposition.

5.3 Analytical Application

The framework introduced here may be applied across a range of research and design contexts. In qualitative analysis, it can serve as a lens for identifying points at which users encounter misalignment between expected and actual system behavior. More specifically, it makes visible when system-side constraints become user-side preprocessing accompanied by interpretive burden, rather than appearing only as generic inefficiency. In evaluation settings, it can inform the development of exploratory instruments that assess perceived understanding, predictive accuracy, interaction confidence, and conceptual transparency. In design contexts, it can guide the reallocation of preprocessing functions back to systems in ways that reduce interpretive burden.

While the present study focuses on document submission workflows, similar dynamics are likely to appear in other domains, including enterprise systems, data-entry environments, and human–AI interaction contexts, where system logic is often opaque and user understanding is too often assumed rather than supported.

6 Implications for Design

Reducing cognitive distance in document submission systems requires rethinking how preprocessing work is distributed between users and systems.

Systems can reduce friction by accepting a broader range of inputs and performing feasible transformations internally. Techniques such as client-side image compression, format normalization, and pre-submission validation can often be implemented at relatively low computational cost.

Where automatic transformation is infeasible, interpretability becomes even more important. Constraints should be presented early, expressed in language aligned with user understanding, and accompanied by actionable guidance. Feedback should clarify not only that an error occurred, but also why it occurred and how it can be resolved.

Terminology also plays a significant role. Labels that presume prior technical knowledge can introduce avoidable barriers. Reframing interaction elements in clearer, task-oriented language can reduce ambiguity and improve interpretive clarity.

7 Research Implications

This conceptual case analysis suggests that cognitive distance is not limited to complex or specialized systems. It can also emerge in routine interactions shaped by commonplace design practices.

Future research should examine how cognitive distance varies across domains, how it changes with repeated use, and how it interacts with factors such as trust, learning, and system adoption. Empirical work would be especially valuable where perceived workload and interpretive understanding diverge, including cases in which users complete tasks successfully while remaining unable to explain the system's behavior.

By grounding cognitive distance in analytically reconstructed interaction patterns, this paper supports its further development as a useful concept for information systems research.

8 Conclusion

Document submission systems show how cognitive distance can be produced through ordinary design decisions. The issue is not simply that such systems require effort; it is that they may require users to reason in terms that reflect system logic rather than their own.

Addressing this problem requires more than interface simplification. It requires aligning system behavior with human reasoning, reallocating appropriate preprocessing functions to systems, and making causal relations within the system more legible to users.

Cognitive distance provides a useful framework for understanding this form of misalignment. This conceptual case analysis does not argue that all document submission systems are opaque or that all upload failures reduce to cognitive distance. Rather, it argues that cognitive distance offers an analytically useful explanation for a recurrent class of interaction breakdowns reconstructed here that conventional workload-based accounts do not fully capture. The contribution is therefore not merely to observe that poor design exists, but to identify a recurring allocation pattern of processing responsibilities that helps explain a specific class of breakdowns. The broader challenge for information systems design is therefore not only to build systems that function, but to build systems whose logic remains understandable to the people expected to use them.

Declaration of generative AI use

The author used generative AI tools to support language refinement and editorial clarity during the preparation of this manuscript. All conceptual development, analysis, and final content were produced and verified by the author.

References

1 Author (2026a).
From cognitive load to cognitive distance: Reframing usability and accessibility in information systems.
[Details omitted for double-blind review].

2 Author (2026b).
Measuring cognitive distance in information systems: Toward an operational framework for usability, accessibility, and conceptual alignment.
[Details omitted for double-blind review].

3 Norman, D. A. (2013).
The design of everyday things (Revised and expanded ed.).
Basic Books.
Publisher page: Hachette Book Group title record.
DOI not listed in the publisher record consulted.

4 Nielsen, J. (1993).
Usability engineering.
Morgan Kaufmann.
DOI: https://doi.org/10.1016/C2009-0-21512-1

5 Jarrett, C., & Gaffney, G. (2009).
Forms that work: Designing web forms for usability.

Morgan Kaufmann.
Google Books bibliographic record: title record.
DOI not listed in the bibliographic record consulted.

6 Sweller, J. (1988).
Cognitive load during problem solving: Effects on learning.
Cognitive Science, 12(2), 257–285.
DOI: https://doi.org/10.1016/0364-0213(88)90023-7

7 British Standards Institution. (2018).
BS EN ISO 9241-11:2018: Ergonomics of human-system interaction—Usability: Definitions and concepts.
BSI. Identical adoption of ISO 9241-11:2018.
DOI: https://doi.org/10.3403/30319071

8 Zaheri, M., Famelis, M., & Syriani, E. (2025).
Workarounds in software forms: A user study.
In *Proceedings of the 2025 IEEE International Conference on Collaborative Advances in Software and Computing (CASCON 2025)* (pp. 439–447).
DOI: https://doi.org/10.1109/CASCON66301.2025.00073